Computer Ethics

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Chapter one

An Overview of Ethics

Basic Definitions

Moral code

- Set of rules
- Establishes boundaries of generally accepted behavior
- Different rules often have contradictions

Morality

- Social conventions about right and wrong
- Widely shared
- Form basis for an established consensus

Basic Definitions

- **Morality may vary by:**
- Age
- Cultural group
- Ethnic background
- Religion
- Life experiences
- Education
- Gender

Ethics

- Set of beliefs about right and wrong behavior

Basic Definitions

Virtues

- Habits that incline people to do what is acceptable

Vices

- Habits of unacceptable behavior

Virtues and vices define a personal value system

- Scheme of moral values

The Importance of Integrity

- Integrity is a cornerstone of ethical behaviour
- People with integrity:

o Act in accordance with a personal code of principles

o Extend to all the same respect and consideration

o Apply the same moral standards in all situations

- Lack of integrity emerges if you apply moral standards differently according to situation or people involved
- Many ethical dilemmas are not as simple as right versus wrong

The Difference between Morals, Ethics, and Laws:

Morals:

- one's personal beliefs about right and wrong

Ethics:

- standards or codes of behavior expected of an individual by a group

Law:

- system of rules that tells us what we can and cannot do
- Laws are enforced by a set of institutions
- Legal acts conform to the law
- Moral acts conform to what an individual believes is the right belief of right and wrong

Ethics in the Business World

- Both the likelihood and the negative impact of inappropriate behavior have increased
- Several trends have increased the likelihood of unethical behaviour:
- Globalization creating complex work environments
- Organizations challenged to maintain profits / revenue
- Heightened vigilance by:
- Employees
- Shareholders
- Regulatory agencies

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Recent scandals in IT companies

- Satyam Computer Services (India)
- Hewlett Packard
- Computer Associates International
- IBM
- Not just executives, but even lower-level employees, can find themselves in the middle of an ethical dilemma

Why Fostering Good Business Ethics Is Important

- To gain the good will of the community
- To create an organization that operates consistently
- To foster good business practices
- To protect organization/employees from legal action
- To avoid un favourable publicity

Gaining the Good Will of the Community

- Organizations have fundamental responsibilities to society
- Declared in formal statement of company's principles or beliefs
- Include:
- Making contributions to charitable organizations and non-profit institutions
- Providing benefits for employees in excess of legal requirements
- Choosing economic opportunities that might be more socially desirable than profitable
- Socially responsible activities create good will
- Good will makes it easier for corporations to conduct business

Creating an organization that operates consistently

Consistency ensures that employees:

- Know what is expected of them
- Can employ the organization's values to help them in decision making

Consistency also means that shareholders, customers, suppliers, and community know what they can expect of the organization

Many companies share the following values:

- Operate with honesty and integrity, staying true to organizational principles
- Operate according to standards of ethical conduct, in words and action
- Treat colleagues, customers, and consumers with respect
- Strive to be the best at what matters to the company 4
- Value diversity
- Make decisions based on facts and principles

Fostering Good Business Practices

- Good ethics means good business/improved profits
- Companies that:
- Produce safe and effective products
  - Avoid costly recalls and lawsuits
- Provide excellent service that retains customers
- Develop and maintain strong employee relations
  - Suffer lower turnover rates
  - Enjoy better employee morale

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- Suppliers/business partners place priority on working with companies that operate in a fair and ethical manner
- Bad ethics means bad business/waning profits
- Bad ethics can lead to bad business results
- Bad ethics can have a negative impact on employees

Protecting the Organization and Its Employees from Legal Actions

- U.S. Supreme Court established that an employer can be held responsible for the acts of its employees
- This principle is called respondent superior
- Coalition of several legal organizations argues establishment of ethics and compliance programs should reduce criminal liability of organization
- Others argue company officers should not be given light sentences if their ethics programs are ineffective

Avoiding Un favourable Publicity

- Public reputation of company strongly influences:
- Value of its stock
- How consumers regard products and services
- Degree of oversight received from government
- Amount of support and cooperation received
- Organizations are motivated to build strong ethics programs to avoid negative publicity

Improving Corporate Ethics

Characteristics of a successful ethics program

- Employees willing to seek advice about ethical issues
- Employees feel prepared to handle situations that could lead to misconduct
- Employees are rewarded for ethical behavior
- Employees are not rewarded for success obtained through questionable means
- Employees feel positive about their company

Ethical Standards Set by Board of Directors

Board oversees the organization's business activities and management

Board members of company are expected to:

- Conduct themselves according to the highest standards of personal and professional integrity
- Set standard for company-wide ethical conduct
- Ensure compliance with laws and regulations
- Create environment in which employees can seek advice about business conduct, raise issues, and report misconduct

Establishing a Corporate Code of Ethics
Code of ethics

- Highlights an organization's key ethical issues
- Identifies overarching values and important principles
- Focuses employees on areas of ethical risk
- Offers guidance for employees to recognize and deal with ethical issues
- Provides mechanisms to report unethical conduct
- Help employees abide by the law, follow necessary regulations, and behave in an ethical manner

Sarbanes-Oxley Act of 2002

- Enacted in response to public outrage over several major accounting scandals
- Section 404 requires that the Chief Executive Officer (CEO) and Chief Financial Officer (CFO) sign any Securities and Exchange Commission (SEC) filing to attest to its accuracy
- Section 406 requires public companies to disclose whether or not they have a code of ethics and if any waivers to that code have been granted
- Cannot gain company-wide acceptance unless it is:
  - Developed with employee participation
  - Fully endorsed by organization's leadership
- Must continually be applied to company's decision making and emphasized as part of its culture
- Breaches in the code of ethics must be identified and dealt with appropriately

Conducting Social Audits
Social audit

- Reviews how well organization is meeting ethical and social responsibility goals
- Communicates new goals for upcoming year
- Shared broadly with employees, shareholders, investors, market analysts, customers, suppliers, government agencies, and local communities

Requiring Employees to Take Ethics Training

- Personal convictions improved through education
- Comprehensive ethics education program encourages employees to act responsibly and ethically
  - Often presented in small workshop formats
  - Employees apply code of ethics to hypothetical but realistic case studies
  - Demonstration of recent company decisions based on principles from the code of ethics
- Critical that training increase the percentage of employees who report incidents of misconduct
- Employees must:
  - Learn effective ways of reporting incidents
  - Be reassured their feedback will be acted on without retaliation

Including Ethical Criteria in Employee Appraisals

- Only 43% of companies include ethical conduct in employee's performance appraisal
- Ethical criteria include:

o Treating others fairly and with respect

o Operating effectively in a multicultural environment

o Accepting personal accountability

o Continually developing themselves and others

o Operating openly and honestly with all

Creating an Ethical Work Environment

- Good employees may make bad ethical choices
- May be encouraged to do “whatever it takes” to get the job done
- Employees need a knowledgeable resource to discuss perceived unethical practices

o A manager

o Legal or Internal Audit Department

o Business Unit’s legal counsel

o Anonymously through internal Web site

Including Ethical Considerations in Decision Making

Steps in a decision-making process

– Develop problem statement

– Identify alternatives

– Evaluate and choose alternative

– Implement decision

– Evaluate results

– Success

Develop a Problem Statement

- Clear, concise description of the issue
- Answers these questions:
  - What causes people to think there is a problem?
  - Who is directly affected by the problem?
  - Is there anyone else affected?
  - How often does it occur?
  - What is the impact of the problem?
  - How serious is the problem?
- Most critical step in decision-making process

Example of a good problem statement:

– “Our product supply organization is continually running out of stock of finished products, creating an out-of-stock situation on over 15 percent of our customer orders, resulting in over $300,000 in lost sales per month.”

Examples of poor problem statements:

– “We need to implement a new inventory control system.” (possible solution, not a problem statement)

– “We have a problem with finished product inventory.” (not specific enough)

Identify, Evaluate, and Choose an Alternative

- Enlist help to brainstorm alternative solutions
- Evaluate by weighing laws, guidelines, and principles
- Consider likely consequences of each alternative
- Alternative selected must:
  - Be ethically and legally defensible
  - Be consistent with policies and code of ethics
  - Take into account impact on others
  - Provide a good solution to problem

Common Approaches to Ethical Decision Making

Virtue ethics approach

– Focuses on concern with daily life in a community

– People guided by virtues to reach “right” decision

– More effective than following set of principles/rules

Problems

– Does not provide guide for action

– Virtue cannot be worked out objectively; depends on circumstances

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Utilitarian Approach

– Chooses action that has best overall consequences

– Finds the greatest good by balancing all interests

– Fits concept of value in economics and the use of cost-benefit analysis

Problems

– Measuring and comparing values is often difficult

– Predicting resulting benefits and harm is difficult

Fairness Approach

– Focuses on fair distribution of benefits/burdens

– Guiding principle is to treat all people the same

Problems

– Decisions can be influenced by personal bias

– Others may consider the decision unfair

Common Good Approach

– Work together for common set of values and goals
– Implement systems that benefit all people
• Problems
– Consensus is difficult
– Some required to bear greater costs than others

Implement the Decision and Evaluate the Results

• **Implement the decision**
– Efficient, effective, timely implementation
– Communication is key for people to accept change
– Transition plan made easy and pain-free
• **Evaluate the results**
– Monitor results for desired effect
– Observe impact on organization and stakeholders
– Return to “Develop problem statement” step if further refinements may be needed

Ethics in Information Technology

- Public concern about the ethical use of information technology includes:
  - E-mail and Internet access monitoring
  - Downloading in violation of copyright laws
  - Unsolicited e-mail (spam)
  - Hackers and identify theft
  - Students and plagiarism
  - Cookies and spyware
- The general public does not understand the critical importance of ethics as applied to IT
- Important decisions are often left to technical experts
- General business managers must assume greater responsibility for these decisions by:
  - Making decisions based on technical savvy, business know-how, and a sense of ethics
  - Creating an environment where ethical dilemmas can be discussed openly, objectively, and constructively

Goals of this text

- – To educate people about the tremendous impact of ethical issues in the successful and secure use of information technology
- – To motivate people to recognize these issues when making business decisions
- – To provide tools, approaches, and useful insights for making ethical decisions

Chapter Two

Accountability in IT

1. Introduction

- Accountability, then, is the means of ensuring that, despite the complexity of the work and the context in which it exists, individuals, groups and, ultimately, the organization as a whole will understand their part in accomplishing goals.
- Accountability is the flip side of delegation.

2. Ensuring Accountability

- Computer systems are powerful and can cause harm financially, physically or psychologically.
- To be mindful of this is one of the professional duties identified in most codes of ethics and/or conduct. One approach would be to find appropriate laws and to use them.
- However, different laws applies in different situations for example, One law might be applicable defective product, while others to negligence. Another factor is the rate of change the technology.
- Typically it can take a long time to pass a piece of legislation, by which time the technology might have evolved.
- In addition to relying on legislation, we can also employ a mix approach in terms of accountability, responsibility, liability and blame. We will discuss these terms individually:

Continue

a. Accountability

- This is used in its broadest meaning. It refers to the appropriate agent to response and depends on various factors. For example, in a department in a company, the head of department might be held accountable. Accountability rests with someone with the ultimate responsibility.

b. Responsibility

- There are many types of responsibility:

 Role Responsibility:

- This is analogous to duty. It is what people are expected to do within their role. For example, in Scenario 1 with LamdaMoo, players are expected to play by the rules, even if they are not stated.
- Most players assume that it means not taking over another players' character.

 Casual Responsibility:

- This is a responsibility as a result of causality. For example, X did something and caused an event to happen. Kim invests a large amount of money in the Market and cause the company to lose money, even though Kim might have done all that was required of her duty.

Continue

Blameworthy Responsibility:

- Kim may or may not be responsible, but she is not blameworthy. Perhaps the software was faulty – in that case, the software designer might be blameworthy if s/he failed to fulfil a role responsibility.

c. Liability

- A friend slips on the polished floor of your house and break a leg. You might be held liable but you may not be blameworthy.

3. Buying and Selling Software

- What are the various responsibilities of the different role players – buyer and seller? The seller:

- Has the right to sell
- Has a duty to be honest
- Has a duty not to coerce a client
- Expected to emphasize the good aspects of the product
- Expected to answer questions honestly

Continue

- Is a seller being dishonest if they do not disclose a problem area not asked about? This is probably the case but it is very difficult to say especially if the user's needs are varied and complex – how does the salesman know what the needs are. To a naïve user, the answer might be yes, but it is difficult to compel this.
- Usually, the product itself (either packaging or licence agreement) will contain all the necessary information. Contract can be voided if all relevant information is not given.
- The buyer also has the responsible to find out all the necessary information and ask the right questions if there is lack of it.

Continue

a. Software – Product or Service?

- This depends on the circumstances of how the software is produced and/or sold.
- A comparison can be made between software and buying a suit. (Prince 1980). There are three types:

• Off the peg / Ready to wear (no alteration)

• Tailor made to specific requirements of person

• In between (oo the peg + alterations)

For software a similar groupings can be made:

• Mass market (a product)

• Customised software (service)

• Mixed (Product + service)

Continue

b.Mass Market

- Products With mass market, strict liability can be imposed on this type of product. Producers can be sued for errors or malfunctions causing damage because:

• Producers puts the product in the public domain and invites people to buy or use it 68

• Producer earns profit and should bear the risks

• Producer is in the best position to anticipate and control the risks associated with the product, thus the onus is on the producer to get the product right.

• Producer can spread the cost of injury and insurance over all clients.

The reason for this utilitarian is that placing the onus on the producer has positive effects.

c. Customized Software

- Strict liability does not make sense here as software created and designed specifically for client. The client knows the context in which the software will be used and so specifies what is required. The client thus needs to take part of the risk. This type of software should thus be considered as a service.

Continue

d. Mixed Case

- The product and service should be treated each on its own.
- **e. Negligence**
- Negligence is a failure to do something that is expected of a reasonable and prudent person.
- For example if a security guard is knocked unconscious, he cannot be blame, but if he is drunk, he might be considered negligent. There is always presumption of a reasonable standard of behavior.
- Often this is used to describe blameworthy behavior of professionals.
- The definition of standard applicable to the offence is often quite difficult to arrive at. Often this is best judged by other professionals (conflict of interest issues?).

Continue

Who was responsible?

- 1970: 2 byte data for storing year information saved expensive space – the professionals did a good job.
- 1980: Same answer
- 1990: Same answer? Were the systems expected to last for 10 years? What is the cost of storage at this time? What is the cost of upgrading old system?
- 1995: Same answer? Were the systems expected to last for 5 years? Storage and upgrade cost? Should professionals warn companies of the Y2K problems? Should CSSA have a recommended code of Practice concerning this issue?
- 1998: Same question, different answer? Client companies all had policies. These were communicated to the shareholders and everyone accepted their responsibilities.
- So it has been a Chapter. The best strategy is:
- • State clearly the role responsibilities of all concerned. Professional needed to explain and document the problem and then to offer options.

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• Make sure that all involved understood the effect of their work on humans. Clients needed to understand available options – they needed to realize that the option of staying with 2 digits was short-term and unsafe.

• Hold those responsible who fail to live up to their responsibility.

f. Diffusion of Responsibility

- in creating a computer system appreciate the following:

• Scale and complexity of the system

• The number of people involved in development, distribution, training and using it.

• What the system will be involved in – many will be involved in important decision making.

• Kant: Humans are responsible for their actions because they have the capacity to control their action.

Example:

- THERAC-25 THERAC-25 is a computer controlled system that gives radiation treatments to patients. Several patients received massive doses, resulting in at least three fatalities.
- It was difficult to discover who was responsible, but Therac was liable and paid compensation to the victims' families.
- Eventually it was found that the action of the operator has caused the accidents. However, it was also found that if an operator entered incorrect mode, noticed this and corrected the error (within 8 seconds as specified), the system still went wrong and the patient was killed as a result. Thus while the operator caused the accident they were not to blame.

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- The error was traced to also exist in previous version of the software, but had caused no problems because there was only one mode in that version. Was it the designer or the tester (who looks for errors and unconformities to the requirements) who was at fault? The Therac case also illustrates the difficulty of testing real-time systems. Should there have been a feature on the system limiting the radiation dose to some MAXIMUM irrespective of everything else. If so, should this not have been specified by the clients? Are they also partially to blame?
- **Example:**
- ISP Responsibility Recall the scenario with Milo, a service provider provides him with chat rooms and forums facilities. Milo was defamed by another user and has the right of recourse. The obvious route to deal with this is the hold the
- Individual responsible – the problem is that the individual is anonymous.
- Stratton versus ISP Prodigy (1995) Prodigy was sued by users with respect to content in its online forums which Prodigy has advertised that it has editorial control over. The court has applied the law governing newspapers and found Prodigy guilty. However, Prodigy claimed that they were like a telephone company and not newspaper company. The following year, the US Communications Decency Act reversed this on the basis that ISPs are closer to telephone companies. The act also introduced the Good Samaritan immunity in which an entity can exercise control without liability and is encouraged to do so.

Chapter Three

Ethics for IT Workers and IT Users

IT Professionals

Profession is a calling that requires:

- Specialized knowledge
- Long and intensive academic preparation
- Professionals:
- Require advanced training and experience
- Must exercise discretion and judgment in their work
- Their work cannot be standardized
- Contribute to society, participate in lifelong training, assist other professionals
- Carry special rights and responsibilities

IT Workers Professionals

Partial list of IT specialists

- Programmers
- Systems analysts
- Software engineers
- Database administrators
- Local area network (LAN) administrators
- Chief information officers (CIOs)

Legal perspective

- IT workers do not meet legal definition of professional
- Not licensed by state or federal government
- Not liable for malpractice

The Changing Professional Services Industry

- IT workers are considered part of the professional services industry
- Seven forces are changing professional services
- Client sophistication (able to drive hard bargains)
- Governance (due to major scandals)
- Connectivity (instant communications)
- Transparency (view work-in-progress in real-time)
- Modularization (able to outsource modules)
- Globalization (worldwide sourcing)
- Commoditization (for low-end services)

Professional relationships that must be managed

IT workers involved in relationships with:

- Employers
- Clients
- Suppliers
- Other professionals
- IT users
- Society at large

Professional relationships that must be managed

IT workers involved in relationships with:

1. Relationships between IT workers and employers
2. Relationships between IT workers and clients
3. Relationships between IT workers and suppliers
4. Relationships between IT workers and other professionals
5. Relationships between IT workers and it users

1. Relationships between IT workers and employers

- IT workers agree on many aspects of work relationship before workers accept job offer
- Other aspects of work relationship defined in company's policy and procedure manual or code of conduct
- Some aspects develop over time
- As steward of organization's IT resources, IT workers must set an example and enforce policies regarding the ethical use of IT in:

Software piracy

- Act of illegally making copies of software or enabling access to software to which they are not entitled
- Area in which IT workers can be tempted to violate laws and policies
- The **Business Software Alliance** (**BSA**) is a trade group representing the world's largest software and hardware manufacturers; mission is to stop the unauthorized copying of software
- Thousands of cases prosecuted each year

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Trade secrets

- Business information generally unknown to public
- Company takes actions to keep confidential
- Require cost or effort to develop
- Have some degree of uniqueness or novelty

Whistle-blowing

- Employee attracts attention to a negligent, illegal, unethical, abusive, or dangerous act that threatens the public interest

2. Relationships between IT workers and clients

IT worker provides:

• Hardware, software, or services at a certain cost and within a given time frame

•**Client provides:**

• Compensation

• Access to key contacts

• Work space

• Relationship is usually documented in contractual terms

• Client makes decisions about a project based on information, alternatives, and recommendations provided by the IT worker

• Client trusts IT worker to act in client's best interests

• IT worker trusts that client will provide relevant information, listen to and understand what the IT worker says, ask questions to understand impact of key decisions, and use the information to make wise choices

• Ethical problems arise if a company recommends its own products and services to remedy problems they have detected

• Creates a conflict of interest

• Problems arise during a project if IT workers are unable to provide full and accurate reporting of a project's status

• Finger pointing and heated discussions can ensue

2. Relationships between IT workers and clients

Fraud

• Crime of obtaining goods, services, or property through deception or trickery

Misrepresentation

• Misstatement or incomplete statement of material fact

• If misrepresentation causes a party to enter into a contract, that party may have the right to cancel contract or seek reimbursement for damages

• Breach of contract

• One party fails to meet the terms of a contract

• When there is material breach of contract:

• The non-breaching party may rescind the contract, seek restitution of any compensation paid to the breaching party, and be discharged from any further performance under the contract

IT projects are joint efforts in which vendors and customers work together

• When there are problems, it is difficult to assign who is at fault

3. Relationships between IT workers and suppliers

Develop good working relationships with suppliers:

- To encourage flow of useful information and ideas to develop innovative and cost-effective ways of using the supplier in ways that the IT worker may not have considered
- By dealing fairly with them
- By not making unreasonable demands

Bribery

- Providing money, property, or favors to obtain a business advantage
- U.S. Foreign Corrupt Practices Act (FCPA): crime to bribe a foreign official, a foreign political party official, or a candidate for foreign political office
- At what point does a gift become a bribe?
- No gift should be hidden
- Perceptions of donor and recipient can differ
- United Nations Convention against Corruption is a global treaty to fight bribery and corruption

4. Relationships between IT workers and other professionals

- Professionals feel a degree of loyalty to other members of their profession
- Professionals owe each other adherence to their profession's code of conduct
- Ethical problems among the IT profession
- Résumé inflation on 30% of U.S. job applications
- Inappropriate sharing of corporate information
- Information might be sold intentionally or shared informally with those who have no need to know

5. Relationships between IT workers and it users

IT user:

- Person using a hardware or software product
- IT workers' duties
- Understand users' needs and capabilities
- Deliver products and services that meet those needs
- Establish environment that supports ethical behaviour:
- To discourages software piracy
- To minimize inappropriate use of corporate computing resources
- To avoid inappropriate sharing of information

6. **Relationships between IT workers and society**

Society expects members of a profession:

- To provide significant benefits
- To not cause harm through their actions
- Actions of an IT worker can affect society
- Professional organizations provide codes of ethics to guide IT workers' actions